Ivan Vashchenko

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I own projects end-to-end, from requirements and architecture to deployment, while leading cross-functional work at WTI and running student engineering initiatives as Applied Computational Modeling Club Founder and President.

Education

Bachelor of Science in **Computer Science and Mechanical Engineering |** Principia College | May 2027
Bachelor of Science in **Mechanical Engineering |** University of North Dakota | May 2027
GPA: 3.5

Software Architecture | Programming Languages | Machine Component Design | Manufacturing Processes | Finite Element Analysis
Computational Fluid Dynamics | Electrical Engineering Circuits | Statistics | Design of Machinery | Data Structures | Aerodynamics

Professional Experience

Mechanical Engineering Intern | WTI Services Pure Air | Clearwater, FL & Remote| May 2024 – Present

- **Built a 3D reconstruction and visualization workflow** using **photogrammetry and parametric modeling** to reason about large mechanical assemblies.
- **Integrated 3D visualization and interaction tools** to compare existing vs. proposed configurations, **improving system-level decision making.**
- Developed internal engineering tools that **automated data pipelines** and reduced manual handling, **improving turnaround by 40–45%.**
- **Worked independently** on long-term projects and **collaborated with engineering leadership** to scale modeling and **automation tools into CAD/BIM workflows**.

Projects

Autonomous Sorting & Packaging System — Mechanical Integration Project

Designed and built a **robotic sorting system** where **camera-based ML classification** drives physical actuation.

- Integrated **Raspberry Pi vision pipeline**, image preprocessing, and inference with **servo-controlled mechanical subsystems.**
- Iterated on sensing setup, timing, and **mechanical tolerances** to improve reliability and repeatability.

AHU Digitalization Pipeline (End-to-End CV/3D Automation)

Built an end-to-end pipeline that converts **iPhone field video → photogrammetry 3D reconstruction → structured deliverables** (interactive 3D viewer + 2D CAD sections + component metadata) for Sales and Engineering.

- Automated reconstruction processing (scale/orientation consistency via control points), generated 2D section outputs, and implemented **DXF cleanup/orthogonalization using a C# DXF library** for AutoCAD-ready geometry.
- Developed **ML-driven component recognition** workflows (classification + YOLO-based detection) to localize components for downstream documentation and automation.

ML Signal Extraction from Unstructured Audio Data – Hackathon Project

Built an **ML pipeline to extract structured signal from noisy, unstructured audio data using Librosa** and neural models (**YourTTS**).

- Focused on preprocessing, feature extraction, and model behavior under imperfect real-world inputs.

QuickScope Estimation Tool – Engineering Systems Tooling

- Built automation platform with **C#, VBA, Salesforce, and Google Maps API** to handle **multi-department project estimation.** Automated **metadata handling** and reporting features with a user-friendly interface.

Technical Skills

Languages: Python | C++ | C# | MATLAB | Java | JavaScript
Robotics & Perception: Camera-based sensing | ML-assisted classification | photogrammetry | point clouds | LiDAR
3D & Simulation: Geometric modeling |parametric systems |digital twin workflows |Three.js
Systems & Infrastructure: Git | hardware–software integration |sensor pipelines
Hardware & Prototyping: Raspberry Pi | servo actuation | mechanical system integration

Leadership

Founder and President | Applied Computational Modeling Club (ACMC) | 2025 – Present

- Student organization focused on applying computational modeling to real-world engineering problems.

House President | Principia College | 2024 – Present

- Coordinating operations, budgeting, and communication for a residential dorm community.

President | Christian Science Organization | 2025 – Present

- Organization of 100+ members, coordinating events cross-functional teams, fostering an inclusive community.

The National Society of Leadership and Success (NSLS) | 2025